MILITARY SPECIFICATION

PILLOW, PASSENGER, HEADREST, AIRLINE TYPE

1. SCOPE

1.1 Scope. This specification covers the general and detail requirements for fiber-filled airline-type head rest pillows.

1.2 Classification. Unless otherwise specified, the size shall be as indicated in Table I (see 3.3).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

V-T-276	Thread, Cotton
CCC-C-446	Cloth, Muslin, Cotton

STANDARDS

Federal

FED-STD-191	Textile Test Methods
FED-STD-751	Stitches, Seams, Stitchings

Military

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-129	Marking For Shipment And Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. The requirements for the first article shall be as specified by the procuring activity (see 6.2). The first article will be examined for compliance with the requirements of this specification (see 4.2). The approval of the first article does not relieve the contractor of responsibility for compliance with all other applicable provisions of this specification. When the first article is approved, it shall be used as a guide throughout production of the contract.

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3.2 Materials.

3.2.1 Sheeting. The sheeting for the pillow cases shall be bleached preshrunk cotton muslin and shall conform to Type III, class 3 of CCC-C-446.

3.2.2 Fiber filling. The filling shall be new (not previously used) synthetic fibers consisting of 100 percent polyester fibers manufactured from copolymers of ethylene glycol and terephthalic acid. The staple (denier and length of filaments) of the polyester fibers shall be suitable for meeting the requirements of this specification. Colored, stained and oily fibers will not be acceptable. The fiber filling shall be in batt formation prior to inserting into the pillow cases.

3.2.3 Thread. The thread used for sewing shall conform to the requirements of Type 1A2, 3 cord, Ticket No. 50 (machine thread, bleached) of V-T-276.

3.3 Size and weight. The pillows shall be rectangular in shape without risers or boxes and shall have the particular finished dimensions and weight listed in Table I.

TABLE I - Pillow Size and Weight

Overall Width, Inches *	Overall Length, Inches *	Total (Gross) Weight	
		Min.	Max.
11	15	5.75 oz.	6.25 oz.
*Tolerance in inches:	Width - minus 0, plus 1/2 Length - minus 0, plus 1		

3.3.1 Thickness. The pillow thickness shall be not less than 2-3/4 inches when measured in accordance with 4.4.3.

3.4 Construction. The case of the pillows shall be made of one piece material which shall be folded in half and sewed inside out along one edge in the length dimension and one edge in the width dimension of the pillow, leaving one end in the width dimension of the case open. Stitching shall be not less than 1/4 inch from the edges of the material except on the open end which shall be not less than 1/8 inch from the edge.

3.5 Stitches, seams, and stitching. Stitches, seams and stitching shall conform to the requirements of FED-STD-751. All stitches shall be type 301 with not less than 10 or more than 15 stitches per inch. All seams, except closing seam, shall be type SSa-1. The closing seam shall be type SSc-1 with the turn-under being a minimum of 1/4 inch. All ends of stitching shall be securely backstitched a minimum of 1/2 inch when not caught in another seam or row of stitching.

3.6 Pillow filling. Underweight pillow batts shall not be "pieced in" nor shall "dog ears" be used to fill out the pillow cases since this practice will result in objectionable lumps. When placing the fiber filling into the cases, a funnel shaped compression chute (with outlet dimensions slightly

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smaller than those of the pillow cases) substantially in accordance with Figure 1 shall be used in order to prevent the formation of lumps, fold overs, creases, et cetera. A mechanical aid, rather than a person's hand, shall be used for pushing the fiber filling through the compression chute into the cases. Under no circumstances should the fiber filling be placed into the cases with a blower attachment.

3.7 Identification marking. The letters "MAC" in black, solid block letters 1-1/4 inches high, plus or minus 1/8 inch, shall be permanently and distinctly placed on each side of the pillow at diametrically opposite ends. The lower portion of the letters "MAC" shall be 3/4 inch, plus or minus 1/8 inch, from the applicable end seam. The letters "MAC" shall be placed equidistant, plus or minus 1/2 inch, from the sides of the pillow. The marking shall be of the dry heat transfer (decalcomania) type and shall show good fastness to laundering when tested in accordance with Method 5610 of FED-STD-191.

3.8 Labels. Each pillow shall have a cloth cleaning instructions label and a content label conforming to the legal requirements of the state in which the pillows are manufactured, if required by the state. The text on the cleaning instruction label shall be detached and developed by the contractor; however, acceptability of the text shall be determined by the contracting officer at the time of preproduction evaluation. The cleaning instructions label size shall be no smaller than 3 inches by 4 inches with the printing in capital letters a minimum of 1/8 inch in height. The label or labels shall be integrally sewn in the end seam of the pillow, equidistant from the sides of the pillow. The printing or marking on the label or labels shall show good fastness to laundering when tested in accordance with Method 5610 of FED-STD-191.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

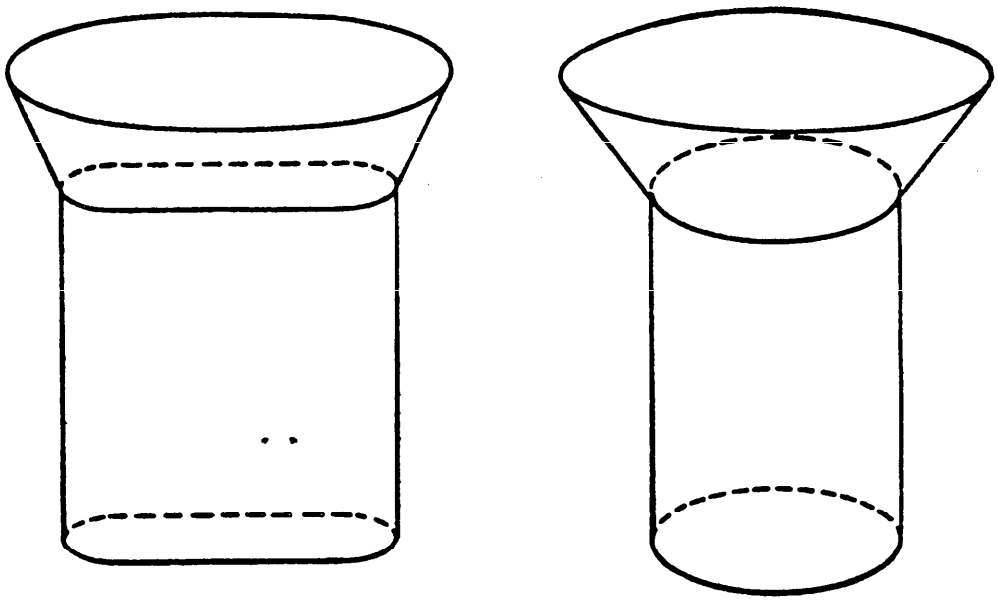
4.2 First article inspection. When required, inspection and testing of the first article sample shall be made of completed pillows for all provisions of this specification applicable to end product examination and tests.

4.3 Inspection for acceptance. Inspection shall be in accordance with the provisions of MIL-STD-105, except where otherwise indicated.

4.3.1 Inspection of components and materials.

4.3.1.1 Testing of components. In addition to the quality assurance testing provisions of the subsidiary specifications and drawings, testing shall be performed conforming to FED-STD-191 on components and material listed in Table II for the characteristics shown therein. The inspection level shall be S-1.

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THE MEASUREMENTS FOR THE SPOUT SECTION ARE GOVERNED BY THOSE OF THE PILLOW SHEETING AND SHOULD BE SUCH THAT THE PILLOW SHEETING FITS ON SNUGLY.

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TABLE II

Component and lot expressed in terms of	Characteristic	Rqmt para	Test Method	Rqmt Appl To Ind Unit	Rqmt Appl To Lot Avg	#Det per smpl Unit	Results rptd as	Smpl unit	AQL
Fiber filling (100 lbs)	Material Identification	3.2.2	Std Com'l	-	X	1	Pass or Fail	1/2 lb	-
Marking (100 heat transfer labels)	Fastness to laundering	3.7	Method 5610	X	-	1	Excellent, good, fair or poor	*	6.5
Labels (100 labels)	Fastness to laundering	3.8	Method 5610	X	-	1 ea type label	Excellent, good, fair or poor	2 imprinted labels (each type)	6.5

* Consistent with each lot or batch of heat transfer labels received, testing shall be performed to determine compliance with the specified performance characteristics. Each sample unit shall be two 3" X 6" swatches of the material being marked bearing the specified marking impression.

4.3.2. In-process inspection. Inspection may be made at any point or during any phase of the manufacturing process to determine whether operation or assemblies are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated nonconformance. In-process inspection shall be conducted to see that accomplishment of the following is in accordance with the specification requirements:

Operation or Assembly	Rqmt Paragraph
Evidence of colored, stained or oily fibers	3.2.2
Construction of pillow case	3.4 and 3.5
Filling of pillow, muslin	3.6

4.3.3 Inspection of the end item.

4.3.3.1 Visual examination. The defects found during examination shall be classified in accordance with 4.3.3.1.1 and 4.3.3.1.2. The lots shall be inspected in accordance with 4.3.3.1.3.

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4.3.3.1.1 Examination for visual characteristics. The sample unit for this examination shall be one completely fabricated pillow. Examination of the pillow for defects in material, workmanship, cleanness, construction and marking:

Examine	Defect	Major	Minor
Quality of muslin material	Any hole, cut or tear	X	
	Any weak place seriously affecting serviceability	X	
	Any weak place affecting serviceability but not seriously		X
	Any smash	X	
	Multiple floats or skips 1/2 inch or more in combined directions	X	
	Multiple floats or skips less than 1/2 inch in combined directions		X
	Three or more contiguous broken or missing warp or filling threads	X	
	Two contiguous broken or missing warp or filling threads		X
	Loose and weakening slub 1/8 inch or more in thickness at widest point	X	
	Slub more than 1/16 inch but less than 1/8 inch thick at widest point		X
	Any spot or stain attributable to the muslin material only, more than 1/4 inch square in area clearly noticeable at a distance of 6 feet		X
	More than two spots or stains attributable to the muslin material only, up to 1/4 inch square in area, clearly noticeable at a distance of 6 feet		X
	Any oil or grease spot or stain attributable to the fiber filling material		X
	Any colored or dyed yarns attributable to the fiber filling material		X

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Examine	Defect	Major	Minor
Workmanship Defects	Damage such as scissor or knife cut, tear, clearly noticeable mend or needle chews, likely to develop into hole	X	
	Evidence of risers or boxes	X	
	Evidence of objectionable lump, fold over or crease	X	
Cleanness	Untrimmed thread ends on exterior of pillow		X
	Fiber filling not removed throughout outside of pillow		X
	Offensive or objectionable odor	X	
Construction	Loose tension resulting in loose seam	X	
	Tight tension (stitches break when normal tension is applied)	X	
	Ends of seams (when not caught in other seams or stitching) not securely backstitched		X
	Seam allowances not maintained, stitched irregularly or unevenly, resulting in badly twisted, pleated or bulging edges		X
	Wrong seam or stitch type		X
	Any open seam caused by one or more broken stitches, or more than two consecutive skipped stitches	X	
	Open seam resulting from 2 consecutive skipped stitches		X

NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken, or when two or more consecutive skipped or runoff stitches occur.

	Raw edge over 1/2 inch in length	X	
	Raw edge over 1/4 inch up to and including 1/2 inch in length		X

NOTE: Raw edges not securely caught in stitching shall be classified as an open seam.

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Examine	Defect	Major	Minor
Construction (Cont'd)	3 or more stitches less than required	X	
	1 or 2 stitches less than required		X
	Covering spliced or not fabricated of one piece of material	X	
	Row of stitching on closing seam less than 1/8 inch from edge		X
Identification markings and labels	Wrong location		X
	Missing, incorrect or illegible		X
	Smeared affecting appearance		X
	Label(s) not attached as specified		X

4.3.3.1.2 Examination for dimensions. Examination shall be made to determine compliance with the dimensional characteristics of the finished pillow. Any dimension that is not within the specified tolerances listed in 3.3 and 3.3.1 shall be classified as a defect. The sample unit for this examination shall be one completely finished pillow. (NOTE: Length and width measurements of the completely filled pillow are taken along seams from end of seam to end of seam.)

4.3.3.1.3 Inspection levels and acceptable quality levels (AQL's). Inspection levels and acceptable quality levels expressed in defects per hundred units shall be as follows:

	Inspection Levels	AQL's
Examination indicated in 4.3.3.1.1	I	Major 2.5 dhu
Examination indicated in 4.3.3.1.1		Total 6.5 dhu
Examination indicated in 4.3.3.1.2	S-3	One class 6.5 dhu

4.3.3.2 Testing of the end item. Testing of the completely fabricated pillow shall be performed in accordance with Table III for the characteristic shown therein. The sample unit shall be one completely fabricated pillow.

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TABLE III

Characteristic	Rqmt Para	Test Method	Rqmt Appl To: Ind Unit	Rqmt Appl To: Lot Avg	#Determ Per Sample Unit	Results Reported As	Insp Level	AQL
Thickness	3.3.1	4.4.3	X	-	1	Avg of 4 measurements to nearest 1/16 inch	S-3	6.5
Weight	3.3	4.4.2	X	-	1	To nearest 0.1 oz	S-3	6.5

4.4 Test methods.

4.4.1 Tests. The various component materials which enter into the fabrication of the pillows shall be tested in accordance with the requirements of this, and other applicable specifications.

4.4.2 Weight. The weight of the pillows shall be determined with a suitable scale capable of weighing the pillows to the nearest 0.1 ounce. In cases of dispute, the gross weight of the pillow shall be determined after exposure for at least 16 hours to an atmospheric condition of 70° to 80° Fahrenheit (F), and 65 ±2 percent relative humidity.

4.4.3 Measurement of thickness. The pillow shall be laid on a clean, flat surface and a square, flat piece of methyl methacrylate plastic, 18 X 18 X 1/4 inches (weighting 3-1/2 pounds ± 1/4 pounds), shall be gently placed on the pillow in a manner that the approximate center of the piece of plastic is over the approximate center of the pillow. The pillow shall not be either fluffed or compressed prior to the foregoing procedure. The thickness shall be taken as the average of the measurements of free height under each of the four corners of the square piece of plastic. The measurements shall be made to the nearest 1/16 inch and taken within one minute after placement of the plastic square on the pillow.

4.4.4 Examination of preparation for delivery requirements. An examination shall be made to determine that packaging, packing, and marking requirements of Section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packed, selected just prior to the closing operation. Defects of closure listed below shall be examined on shipping containers, fully prepared for delivery. The lot size shall be the number of containers in the inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

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Examine	Defects
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location sequence, or method of application.
Materials	Any component missing. Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components, such as: incomplete closure of case liners, container flaps, loose strapping, inadequate stapling. Bulging or distortion of containers.
Content	Number per container is more or less than required.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. Packaging and packing shall be as specified in the Invitation for Bid.

5.2 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The pillows covered by this specification are intended to be used in aircraft as a comfort accessory.

6.2 Order data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. When first article is required (see 3.1).

6.3 Supersession data. Superseding AF Purchase Description 112-SR-72.

Custodian:
Air Force - 82

Review Activity:
Air Force - 82

Preparing Activity:
Air Force - 82

Project No. 7210-F148

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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